



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

Course- CLOUD-NATIVE APPLICATION DEVELOPMENT AND CONTAINER ORCHESTRATION (CNAD)

24CND3101

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TITLE OF THE PROJECT	LogiSphere – Cloud-Native Logistics and Supply Chain Management Platform	
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Abstract:

Modern logistics and supply chain operations involve the movement of products from suppliers to warehouses and customers through activities such as inventory management, order processing, shipment handling, and delivery tracking. Traditional systems may face challenges such as limited coordination, delayed information, inventory inconsistencies, and difficulty in scaling individual operations. These challenges create a need for a flexible and scalable platform that can manage logistics activities efficiently.

To address this need, this project proposes LogiSphere – Cloud-Native Logistics and Supply Chain Management Platform, a digital platform designed to manage and track products, inventory, orders, shipments, and deliveries throughout the supply chain. The system enables suppliers to register products, warehouses to manage inventory, customers to place and track orders, and administrators to monitor logistics operations through a centralized dashboard.

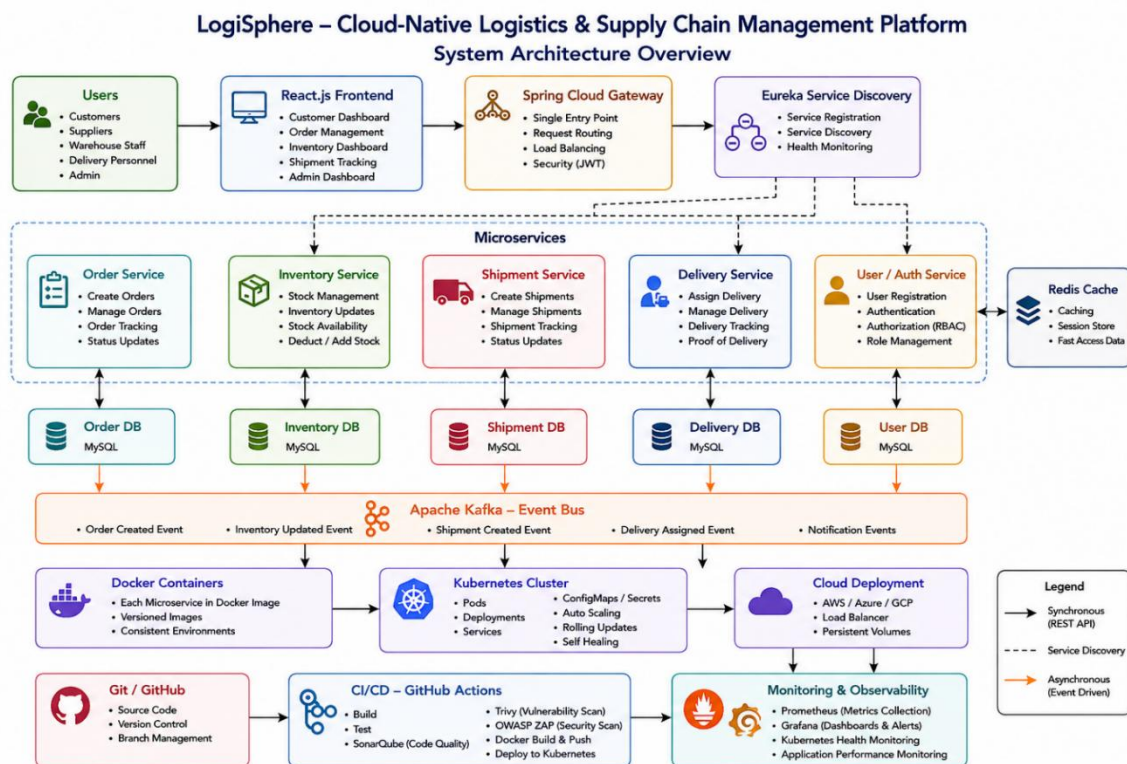
The platform is developed using a microservices-based architecture with independent Spring Boot services such as Order Service, Inventory Service, Shipment Service, Delivery Service, and User/Auth

Service. Spring Cloud Gateway is used for request routing, while Eureka provides service discovery. REST APIs and Apache Kafka enable communication between services, allowing business functions to operate independently and efficiently. MySQL is used for data storage, and React.js provides the web-based user interface.

The cloud-native implementation uses Docker for containerization and Kubernetes for container orchestration, enabling the services to be deployed, managed, and scaled independently. Docker Compose supports local development, while Kubernetes provides deployment, service management, scaling, and recovery capabilities. Git and GitHub support version control and development collaboration, while Prometheus and Grafana can be used for monitoring system performance and service health.

Overall, LogiSphere aims to provide a scalable, reliable, and efficient logistics management solution by combining microservices, APIs, containerization, orchestration, and cloud-native technologies. The platform improves visibility and coordination across the supply chain while demonstrating the practical application of modern cloud-native application development principles.

Project Architecture/Project Overview:



Modern Tools and Technologies

- Java
- Spring Boot
- React.js
- Spring Cloud Gateway
- Eureka
- Apache Kafka
- MySQL
- Redis
- Docker
- Docker Compose
- Kubernetes
- Git & GitHub
- GitHub Actions
- Prometheus
- Grafana
- SonarQube
- Trivy

Conclusion

LogiSphere provides a scalable and reliable cloud-native solution for managing logistics and supply chain operations. By integrating microservices, Docker, Kubernetes, APIs, and cloud-native technologies, the system enables efficient service deployment, scalability, monitoring, and management of logistics activities.